LIPIDS: DERIVED LIPIDS
EICOSANOIDS

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Can you answer following question?

1. What are Eicosanoids?
2. What are different forms of Eicosanoids?
3. How Eicosanoids can be derived?
4. Does their derivation need essential fatty acids?
5. Why Eicosanoids are very important?
6. How Eicosanoids are effecting blood coagulation?

Can you answer following question?

- What are the role of Eicosanoids in pain, inflammation, fever and aspects of reproduction?

- How Eicosanoids are related to Allergic reactions?

Can you answer following question?

- Anti-infilamatory activity of Aspirin is mediated by inhibition of following :

a)Cyclo-oxygenase pathway

b)Lipo-oxigenase pathway

c)Phospholipase A2

d)Epoxides

Can you answer following question?

- Anti-infilamatory activity of steroidal anti infilamatory agents is mediated by inhibition of following :

a)Cyclo-oxygenase pathway

b)Lipo-oxigenase pathway

c)Phospholipase A2

d)Epoxides

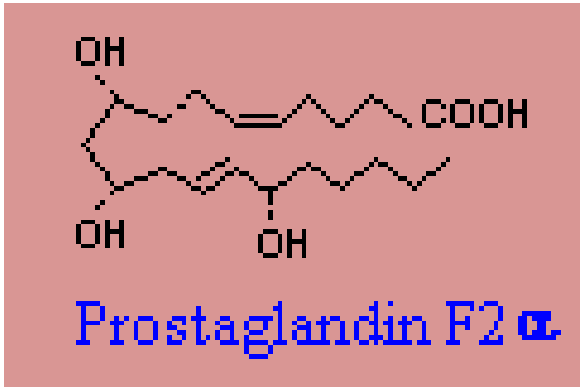
OBJECTIVES

By this interactive lecture students will be able to :

- Enumerate different Eicosanoids
- Differentiate the biochemical activities of different Eicosanoids
- Recognize biochemical importance of Eicosanoids for human life.

Fatty Acid Derivatives - Eicosanoids

- **Eicosanoids** are a large group of molecules derived from polyunsaturated fatty acids.
- The principal groups of hormones of this class are prostaglandins, prostacyclins, leukotrienes and thromboxanes.



Fatty Acid Derivatives - Eicosanoids

- **Arachadonic acid is the most abundant precursor for these hormones.** Stores of arachadonic acid are present in membrane lipids and released through the action of various lipases. The specific eicosanoids synthesized by a cell are dictated by the battery of processing enzymes expressed in that cell.
- These hormones are rapidly inactivated by being metabolized, and are typically active for only a few seconds.

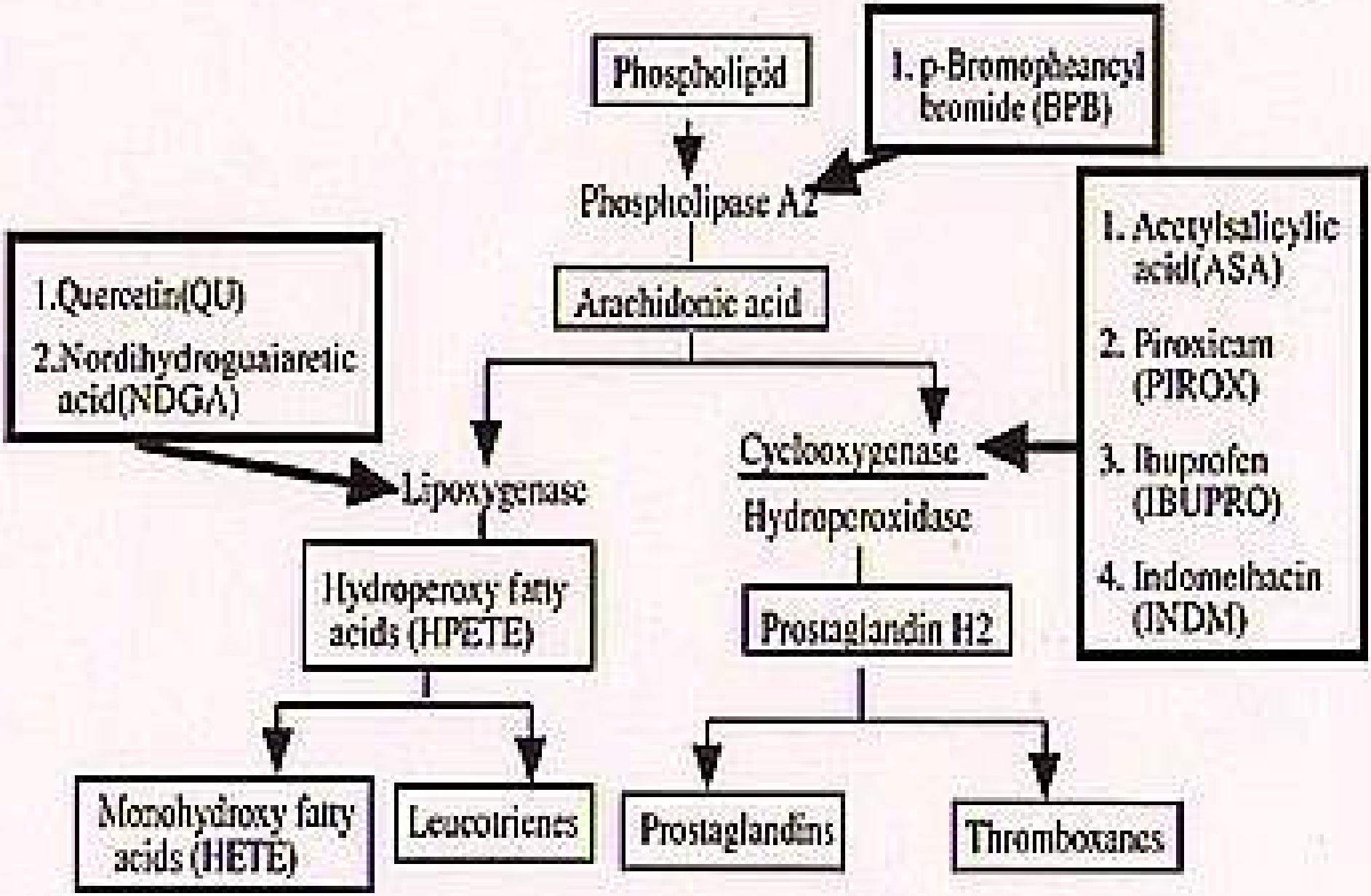
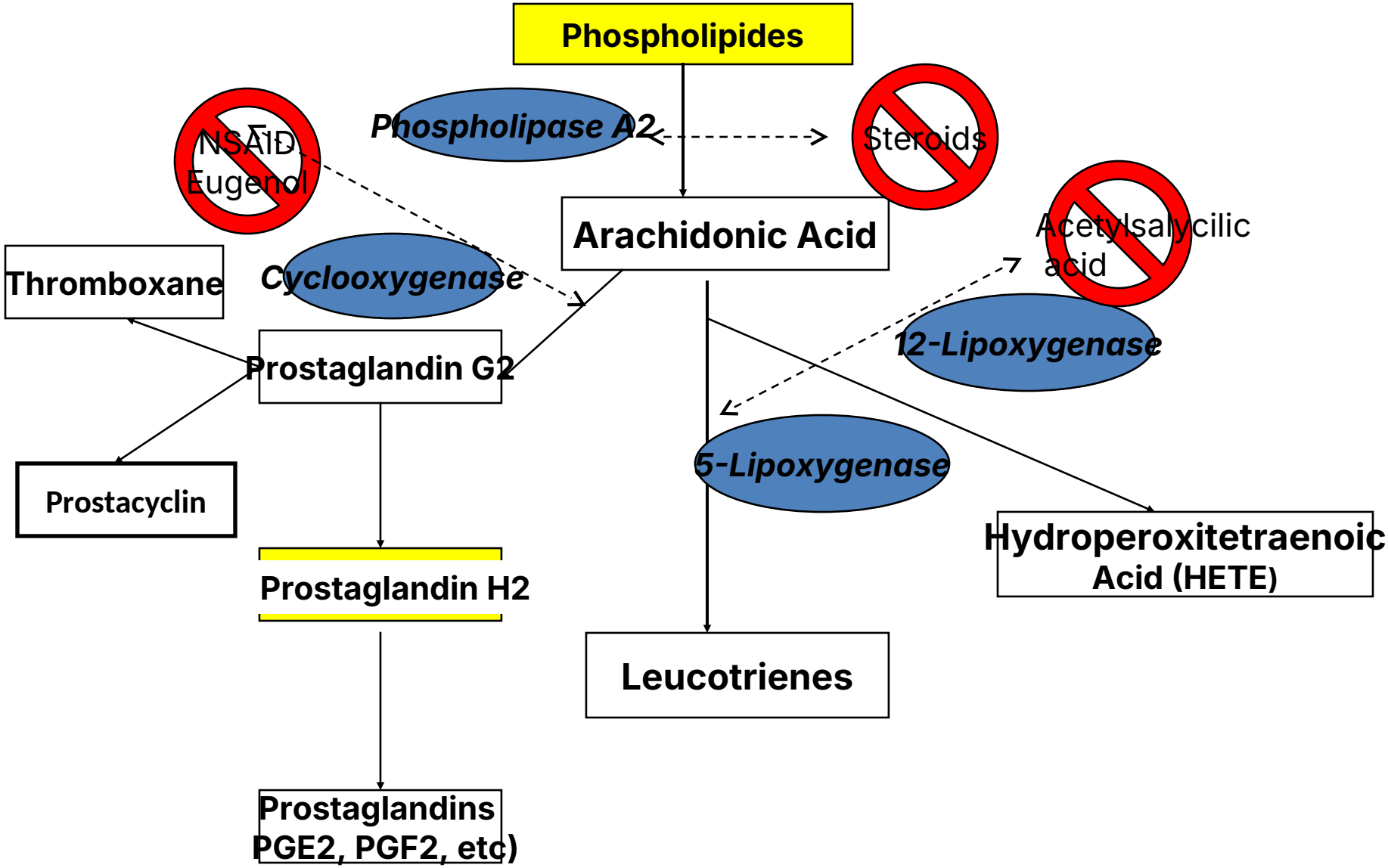


FIG 1. Inhibitors of arachidonic acid cascade.


Phospholipides
Phospholipase A2
NSAID
Eugenol
Steroids
Arachidonic Acid
Acetylsalycilic acid
Thromboxane
Cyclooxygenase
12-Lipoxygenase
Prostaglandin G2
5-Lipoxygenase
Prostacyclin
Hydroperoxitetraenoic Acid (HETE)
Prostaglandin H2
Leucotrienes
Prostaglandins PGE2, PGF2, etc)

BIOCHEMISTRY OF INFLAMMATION

- Phospholipids are the major raw material and starting point for the arachidonic acid pathway. Irritants increase the biosynthesis of phospholipids
- Arachidonic acid is resident to the cell membrane where it is the source of several major biochemical pathways

ARACHIDONIC PATHWAY

- If arachidonic acid is acted upon by cyclo-oxygenase, prostaglandin G2 is generated. It is itself converted into thromboxane or prostacyclin or PGH2, the later then generating the other members of the PG family.
- Thromboxane stimulates platelet aggregation and is a vasoconstrictor

ARACHIDONIC ACID PATHWAY

- Prostacyclin inhibits platelet aggregation and vasoconstriction
- Prostaglandins are non protein chemical mediators: they are fatty acids
- 12-lipoxygenase transforms AA into HETE
- 5-lipoxygenase catalyses the production of the leukotriens (eicosanoid family)

ANTI INFLAMMATORY TESTS

- Cytokines secretion by PBL (human peripheral blood lymphocytes) in culture following addition of a stimulant
- IL 6 release by human fibroblasts

Fatty Acid Modulation

- Reduction in the amount of fat intake enhanced several indices of immune response, including lymphocyte proliferation, natural-killer-cell activity, cytokine production, and delayed-type hypersensitivity
- Intake of omega-3 fatty acids reduced several aspects of neutrophil, monocyte, and lymphocyte functions, including the production of inflammatory mediators

- ***Nutrition*** 2001 Jul-Aug;17(7-8):669-73

Omega 6s as Agents of Inflammation and Decline

- Omega-6 PUFAs , abundant in the Western diet, are precursors for a number of key mediators of inflammation including the 2-series of prostaglandins and IL-6
- PGE_2 , a cyclooxygenase (COX) metabolite of arachidonic acid, a omega-6 PUFA, is a potent mediator of inflammation and cell proliferation
- Omega 6s examples include safflower, corn, soybean, and cottonseed oils

- ***Proc Natl Acad Sci U S A***. 2003 Feb 18;100(4):1751-6

Fatty Acid Modulation
Omega 3s (n-3)

- The parent fatty acid in the omega 3 fatty acid family is alpha-linolenic acid (ALA) which is an essential fatty acid found in high concentrations in certain plant oils, such as flaxseed oil, walnut oil and canola oil
- Omega 3 fatty acids are formed from alpha-linolenic acid and these are mainly found in fish, fish oils and from other marine organisms
- Main marine omega 3 fatty acids are eicosapentaenoic acid (EPA), docosapentaenoic acid and docosahexaenoic acid (DHA)

Fatty Acid Modulation Omega 3s (N-3)

- N-3 Fatty acids favorably affect atherosclerosis, coronary heart disease, inflammatory disease, and perhaps even behavioral disorders
- Docosahexaenoic acid (22:6n-3), which is a vital component of the phospholipids of cellular membranes, especially in the brain and retina, is necessary for their proper functioning

- *Importance of n-3 fatty acids in health and disease*, Connor WE ***Am J Clin Nutr*** 2000 Jan;71(1 Suppl):171S-5S

Fish Oil as Solution

- Increasing the omega-3 content of membrane phospholipid results in a decrease in mitogen-induced PGE(2) synthesis
- Data suggest that replacement of omega-6 PUFA with omega-3 PUFA in cell membranes can result in a decreased cellular response to mitogenic and inflammatory stimuli

- ***Proc Natl Acad Sci U S A***. 2003 Feb 18;100(4):1751-6

How fish oil reduces inflammation:

High prostaglandin E2 and leukotriene B(4) = Inflammation

Fish oil supplements and fish = Reduced arachidonic acid levels

Reduced arachidonic acid = Lower prostaglandin E2 and leukotriene B(4)

Reduced prostaglandin E2 and leukotriene B (4) = Less inflammation

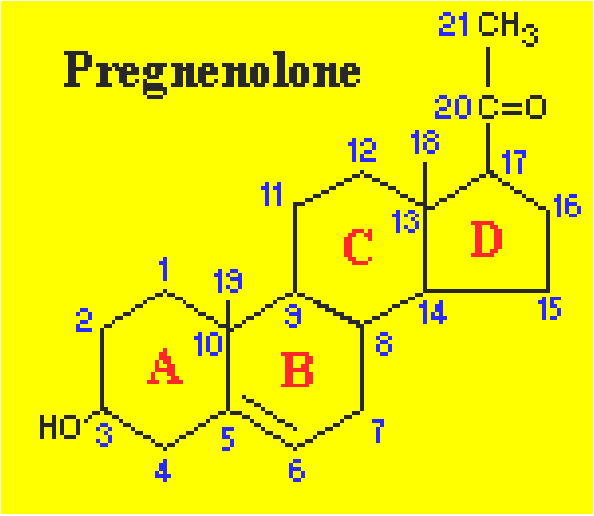


Fish Oil as Healer

- Reduction in strokes
- Decrease in arthritis pain
- Decrease in collagen and bone degrading enzymes
- Modulate inflammation in IBD, autoimmune disorders, dermatitis, asthma,
- Preliminary evidence of cancer prevention (lung, breast, colon)

Steroid hormones

- All steroid hormones are derived from cholesterol and differ only in the ring structure and side chains attached to it.
- All steroid hormones are lipid soluble



Types of steroid hormones

- **Glucocorticoids**; cortisol is the major representative in most mammals
- **Mineralocorticoids**; aldosterone being most prominent
- **Androgens** such as testosterone
- **Estrogens**, including estradiol and estrone
- **Progestogens** (also known a progestins) such as progesterone

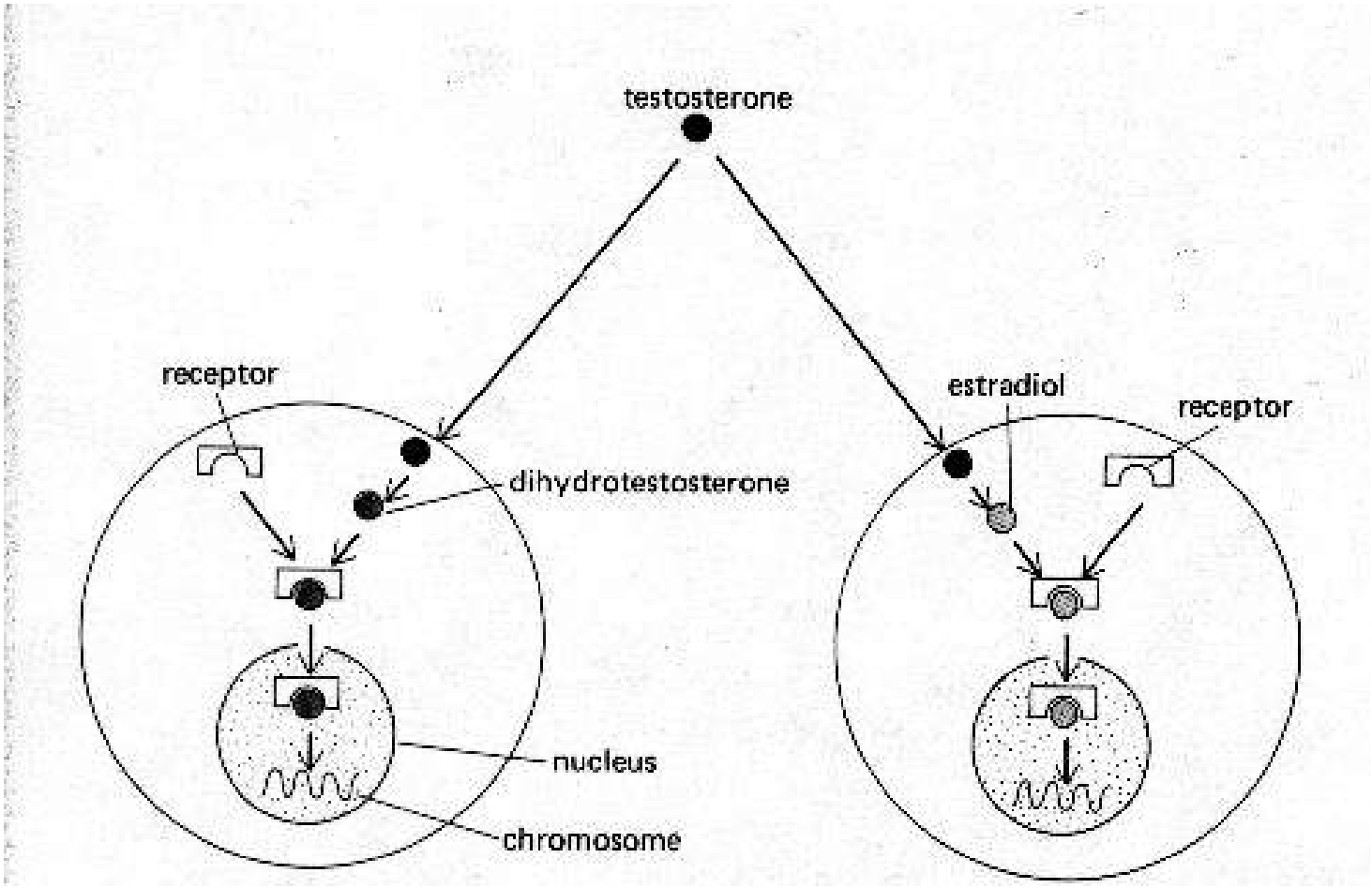
Steroid hormones

- Are not packaged, but synthesized and immediately released
- Are all derived from the same parent compound: Cholesterol
- Enzymes which produce steroid hormones from cholesterol are located in mitochondria and smooth ER
- Steroids are lipid soluble and thus are freely permeable to membranes so are not stored in cells

Steroid hormones

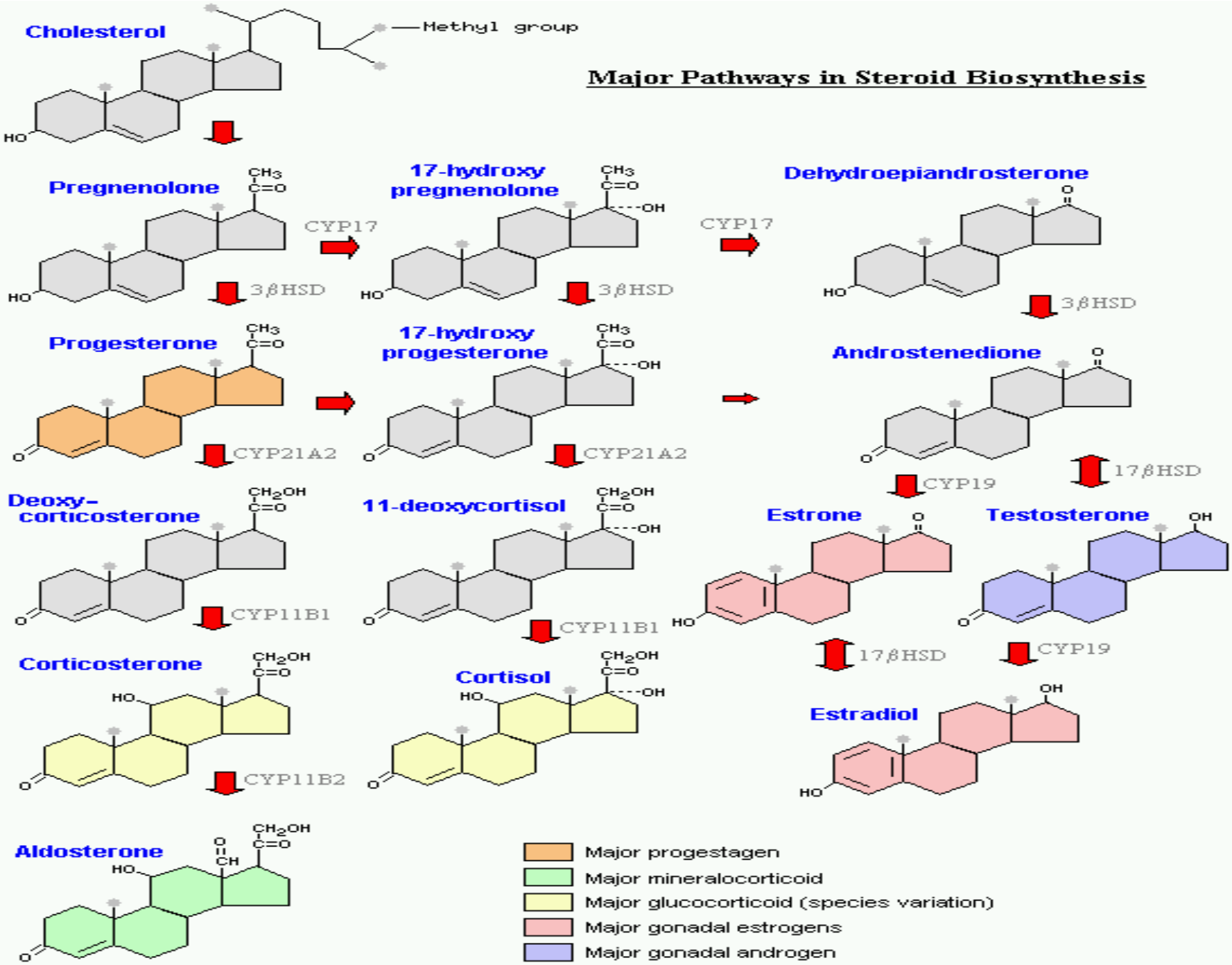
- Steroid hormones are not water soluble so have to be carried in the blood complexed to specific binding globulins.
- Corticosteroid binding globulin carries cortisol
- Sex steroid binding globulin carries testosterone and estradiol
- In some cases a steroid is secreted by one cell and is converted to the active steroid by the target cell: an example is androgen which secreted by the gonad and converted into estrogen in the brain

Steroids can be transformed to active steroid in target cell



Steroidogenic Enzymes

Common name	**"Old" name**	**Current name**
Side-chain cleavage enzyme; desmolase	$P450_{SCC}$	CYP11A1
3 beta-hydroxysteroid dehydrogenase	3 beta-HSD	3 beta-HSD
17 alpha-hydroxylase/17,20 lyase	$P450_{C17}$	CYP17
21-hydroxylase	$P450_{C21}$	CYP21A2
11 beta-hydroxylase	$P450_{C11}$	CYP11B1
Aldosterone synthase	$P450_{C11AS}$	CYP11B2
Aromatase	$P450_{aro}$	CYP19


Cholesterol
Methyl group
Major Pathways in Steroid Biosynthesis
Pregnenolone
CYP17
17-hydroxy pregnenolone
CYP17
Dehydroepiandrosterone
3βHSD
3βHSD
3βHSD
Progesterone
17-hydroxy progesterone
Androstenedione
CYP21A2
CYP21A2
CYP19
17βHSD
Deoxy-corticosterone
11-deoxycortisol
Estrone
Testosterone
CYP11B1
CYP11B1
17βHSD
CYP19
Corticosterone
Cortisol
Estradiol
CYP11B2
Aldosterone
Major progestagen
Major mineralocorticoid
Major glucocorticoid (species variation)
Major gonadal estrogens
Major gonadal androgen

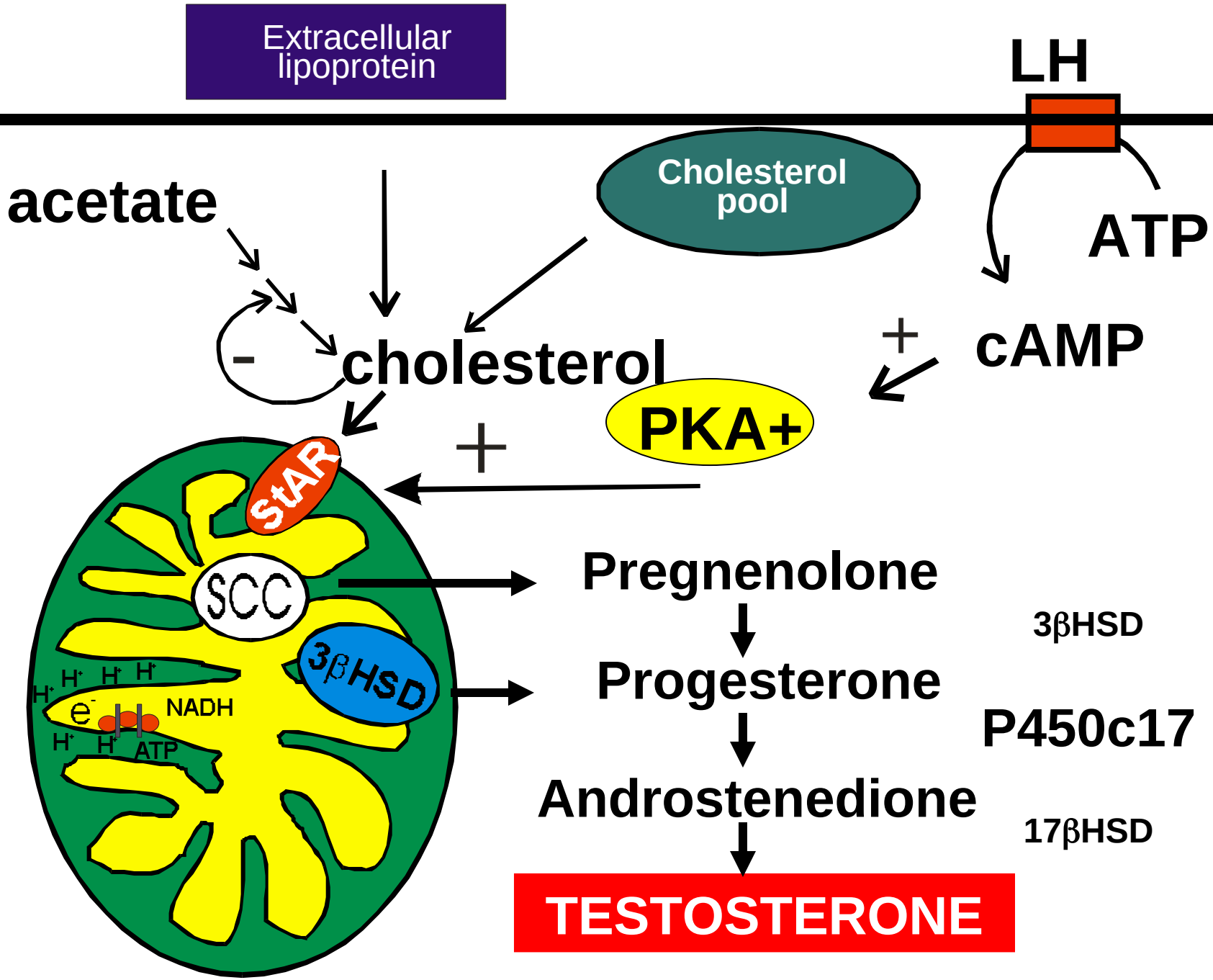
Steroid hormone synthesis

All steroid hormones are derived from cholesterol. A series of enzymatic steps in the mitochondria and ER of steroidogenic tissues convert cholesterol into all of the other steroid hormones and intermediates.

The rate-limiting step in this process is the transport of free cholesterol from the cytoplasm into mitochondria. This step is carried out by the Steroidogenic Acute Regulatory Protein (StAR)

Steroid hormone synthesis

- The cholesterol precursor comes from cholesterol synthesized within the cell from acetate, from cholesterol ester stores in intracellular lipid droplets or from uptake of cholesterol-containing low density lipoproteins.
- Lipoproteins taken up from plasma are most important when steroidogenic cells are chronically stimulated.


Extracellular lipoprotein
LH
ATP
acetate
Cholesterol pool
cAMP
cholesterol
PKA+
StAR
SCC
3βHSD
NADH
ATP
Pregnenolone
3βHSD
Progesterone
P450c17
Androstenedione
17βHSD
TESTOSTERONE

1,25-Dihydroxy Vitamin D3

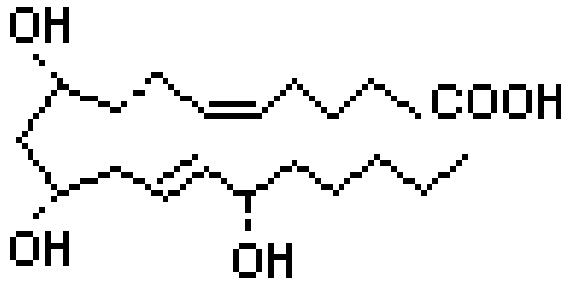
- 1,25-dihydroxy Vitamin D3 is also derived from cholesterol and is lipid soluble
- Not really a “vitamin” as it can be synthesized de novo
- Acts as a true hormone

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Prostaglandin F2α

THANKYOU